

Matrix Computation Homework 4

Yifan Zhang 2025251018 zhangyf52025@shanghaitech.edu.cn

December 11, 2025

Problem 1. (Quadratic Form)

1)

Solution.

It is obviously that:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & a \end{bmatrix}$$

2)

Solution.

First, we find the eigenvalues by solving the characteristic equation $|\mathbf{A} - \lambda \mathbf{I}| = 0$:

$$\det \begin{bmatrix} 2 - \lambda & 1 & 1 \\ 1 & 2 - \lambda & 1 \\ 1 & 1 & a - \lambda \end{bmatrix} = 0$$

It is obviously that:

$$(1 - \lambda)(\lambda^2 - (3 + a)\lambda + 3a - 2) = 0$$

Thus we can get that $\lambda_1 = 1$ and $\lambda_2 + \lambda_3 = 3a - 2$, $\lambda_2 \cdot \lambda_3 = (3 + a)/2$

Since $\mathbf{A}$ is PSD, we need to ensure:

$$\lambda_2 + \lambda_3 \geq 0 \quad \text{and} \quad \lambda_2 \cdot \lambda_3 \geq 0$$

Thus we can get:

$$a \geq \frac{2}{3}$$

3)

Solution.

Since $a = 2$,

$$(1 - \lambda)(\lambda^2 - (3 + a)\lambda + 3a - 2) = 0 \implies \lambda_1 = \lambda_2 = 1, \lambda_3 = 4$$

For $\lambda = 1$, we can easily get

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Apply Gram-schmidt to $\mathbf{w}_1, \mathbf{w}_2$:

$$\begin{aligned} \mathbf{u}_1 &= \mathbf{w}_1 \\ \mathbf{u}_2 &= \mathbf{w}_2 - \frac{\mathbf{w}_2^T \mathbf{u}_1}{\mathbf{u}_1^T \mathbf{u}_1} \mathbf{u}_1 = \begin{bmatrix} 0.5 \\ 0.5 \\ -1 \end{bmatrix} \end{aligned}$$

Normalize the vector and get:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

For $\lambda = 4$, we can easily get

$$\mathbf{q}_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Thus,

$$\begin{aligned} \mathbf{A}^{1/2} &= \sqrt{\lambda_1} \mathbf{q}_1 \mathbf{q}_1^T + \sqrt{\lambda_2} \mathbf{q}_2 \mathbf{q}_2^T + \sqrt{\lambda_3} \mathbf{q}_3 \mathbf{q}_3^T \\ &= 1 \cdot \mathbf{q}_1 \mathbf{q}_1^T + 1 \cdot \mathbf{q}_2 \mathbf{q}_2^T + 2 \cdot \mathbf{q}_3 \mathbf{q}_3^T \\ &= \begin{bmatrix} 4/3 & 1/3 & 1/3 \\ 1/3 & 4/3 & 1/3 \\ 1/3 & 1/3 & 4/3 \end{bmatrix} \end{aligned}$$

Problem 2. (Positive Semidefinite Matrices)

1)

Solution.

Proof symmetric:

$$A^T = (I + \alpha\alpha^T)^T = I^T + (\alpha\alpha^T)^T = I + (\alpha^T)^T\alpha^T = I + \alpha\alpha^T = A$$

Proof PD:

$$\begin{aligned}\mathbf{x}^T \mathbf{A} \mathbf{x} &= \mathbf{x}^T (I + \alpha\alpha^T) \mathbf{x} \\ &= \mathbf{x}^T \mathbf{x} + (\mathbf{x}^T \alpha)(\alpha^T \mathbf{x}) \\ &= \|\mathbf{x}\|_2^2 + \|\alpha^T \mathbf{x}\|_2^2\end{aligned}$$

Thus, when $\mathbf{x} \neq 0$, $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$, i.e. $\mathbf{A}$ is PD matrix.

2)

Solution.

Case 1: The eigenvector is parallel to α

$$\begin{aligned}A\alpha &= (I + \alpha\alpha^T)\alpha \\ &= \alpha + \alpha(\alpha^T\alpha) \\ &= (1 + \beta)\alpha\end{aligned}$$

Thus, $\lambda_1 = 1 + \beta$ is an eigenvalue with eigenvector α . Since α is a single vector, the algebraic multiplicity is 1.

Case 2: The eigenvectors are orthogonal to α

Let v be any non-zero vector satisfy $\alpha^T v = 0$.

$$\begin{aligned}Av &= (I + \alpha\alpha^T)v \\ &= v + \alpha(\alpha^T v) \\ &= v\end{aligned}$$

Thus, $\lambda_2 = 1$ is an eigenvalue.

Since the subspace orthogonal to α has dimension $n - 1$, there are $n - 1$ linearly independent eigenvectors associated with this eigenvalue. The algebraic multiplicity is $n - 1$.

3)

Solution.

It is obviously that:

$$\begin{aligned} C &= (\alpha\alpha^T)^3 \\ &= \beta^2\alpha\alpha^T \\ &= (\beta\alpha)(\beta\alpha)^T \end{aligned}$$

Thus,

$$\begin{aligned} C^{1/2} &= \frac{1}{\|\beta\alpha\|_2}(\beta\alpha)(\beta\alpha)^T \\ &= \frac{\beta^2}{\sqrt{(\beta\alpha)^T(\beta\alpha)}}\alpha\alpha^T \\ &= \sqrt{\beta}\alpha\alpha^T \end{aligned}$$

Problem 3. (Application of Schur Complement)

Solution.

We will use the Schur Complement to solve this problem.

$$M = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix}$$

Then if $A \succ 0$, we just need

$$S = C - B^T A^{-1} B \succeq 0$$

and we will get $M \succeq 0$.

To reduce the computation, we move rows and columns containing a to the end.

Then we can get

$$X = \begin{bmatrix} 2 & \mathbf{2} & 1 & 0 & 0 & -1 \\ \mathbf{2} & \mathbf{a} & \mathbf{2} & \mathbf{0} & \mathbf{0} & \mathbf{2} \\ 1 & \mathbf{2} & 2 & 0 & 0 & 0 \\ 0 & \mathbf{0} & 0 & 2 & 0 & 0 \\ 0 & \mathbf{0} & 0 & 0 & 2 & 0 \\ -1 & \mathbf{2} & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ -1 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 2 & 0 & 0 & 2 \end{bmatrix}^T$$

$$C = a$$

To make X be PSD matrix, we just need to make sure $A \succ 0$ and $S = C - B^T A^{-1} B \succeq 0$.

Step1. Prove $A \succ 0$

Since the 3rd, 4th row of A only have diagonal entry 2, we ignore them to simplify the problem.

Then we focus on

$$A_{core} = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

Since:

- $2 > 0$
- $\det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 3 > 0$
- $\det \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix} = 4 > 0$

we can get that $A \succ 0$.

Step2. Prove $S = C - B^T A^{-1} B \succeq 0$

Let $y = A^{-1} B$, then $Ay = B$.

We can solve the line system and get that:

$$y = [2, 0, 0, 0, 2]^T$$

Thus,

$$\begin{aligned} S &= C - B^T A^{-1} B \\ S &= a - B^T y \\ &= a - \begin{bmatrix} 2 & 2 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix} \\ &= a - 8 \end{aligned}$$

If we want $S \geq 0$, we just need $a \geq 8$

Problem 4. (Matrix Inequality, Matrix Norm)

1)

Solution.

First, we can easily get

$$\sigma_{\min}(\mathbf{X}) = \min_{\|\mathbf{v}\|_2=1} \|\mathbf{X}\mathbf{v}\|_2$$

Since $\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r$ is a real symmetric matrix, we can get:

$$\|\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r\|_2 = \max_{\|\mathbf{v}\|_2=1} |\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r) \mathbf{v}| = \tau$$

Then,

$$\begin{aligned} \max_{\|\mathbf{v}\|_2=1} |\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r) \mathbf{v}| &= \tau \\ |\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r) \mathbf{v}| &\leq \tau \\ |\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} - 1| &\leq \tau \\ \|\mathbf{X}\mathbf{v}\|_2^2 - 1 &\leq \tau \end{aligned}$$

Thus,

$$1 - \tau \leq \|\mathbf{X}\mathbf{v}\|_2^2 \leq 1 + \tau$$

Since $1 - \tau \leq \|\mathbf{X}\mathbf{v}\|_2^2$, we can get:

$$\min_{\|\mathbf{v}\|_2=1} \|\mathbf{X}\mathbf{v}\|_2^2 \geq 1 - \tau$$

i.e.,

$$\sigma_{\min}(\mathbf{X})^2 \geq 1 - \tau$$

$$\sigma_{\min}(\mathbf{X}) \geq \sqrt{1 - \tau}$$

Since $0 \leq \tau < 1$, we know that $\sqrt{1 - \tau} \geq 1 - \tau$.

Therefore:

$$\sigma_{\min}(\mathbf{X}) \geq 1 - \tau$$

2)

Solution.

$$\sigma_1(\mathbf{E}) = \sqrt{\lambda_{\max}(\mathbf{E}^\top \mathbf{E})}$$

First, we get $\mathbf{E}^\top \mathbf{E}$

$$\begin{aligned}\mathbf{E}^\top \mathbf{E} &= (\mathbf{a}\mathbf{b}^\top)^\top (\mathbf{a}\mathbf{b}^\top) \\ &= \|\mathbf{a}\|_2 \mathbf{b}\mathbf{b}^\top\end{aligned}$$

Since $\mathbf{E}^\top \mathbf{E} = \|\mathbf{a}\|_2 \mathbf{b}\mathbf{b}^\top$ is a rank-1 matrix, we can get that $\mathbf{E}^\top \mathbf{E}$ only have one none zero eigenvalue.

Consider the vector $\mathbf{b}$ as a candidate eigenvector for the matrix $\mathbf{b}\mathbf{b}^\top$:

$$(\mathbf{b}\mathbf{b}^\top)\mathbf{b} = \mathbf{b}(\mathbf{b}^\top \mathbf{b}) = \mathbf{b}(\|\mathbf{b}\|_2^2) = \|\mathbf{b}\|_2^2 \mathbf{b}$$

This shows that $\mathbf{b}$ is an eigenvector of $\mathbf{b}\mathbf{b}^\top$ with respect to the eigenvalue $\|\mathbf{b}\|_2^2$.

Thus, the non-zero eigenvalue of the full matrix $\mathbf{E}^\top \mathbf{E} = \|\mathbf{a}\|_2^2 (\mathbf{b}\mathbf{b}^\top)$ is:

$$\lambda_{\max} = \|\mathbf{a}\|_2^2 \cdot \|\mathbf{b}\|_2^2$$

$$\sigma_1(\mathbf{E}) = \sqrt{\lambda_{\max}(\mathbf{E}^\top \mathbf{E})} = \|\mathbf{a}\|_2 \cdot \|\mathbf{b}\|_2$$